MATH 2431 Honors Probability Spring 2026

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# Solutions to Problem Set 5

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1. **($*$) $\sigma$-algebras and random variables** [4 Points]

In a random experiment, a green and a blue die are tossed at the same time. We choose $\Omega = \{1, \dots, 6\}^2$ (with $(\omega_1, \omega_2)$ denoting the numbers shown by the (blue, green) die) and consider the following two $\sigma$-algebras:

$$\begin{aligned} \mathscr{F} &= \mathscr{P}(\Omega), \\ \mathscr{F}_{\mathrm{sym}} &= \{A \subseteq \Omega \,:\, \text{for all } (\omega_1, \omega_2) \in \Omega, (\omega_1, \omega_2) \in A \iff (\omega_2, \omega_1) \in A\}. \end{aligned}$$

(a) Note that $\mathscr{F}_{\mathrm{sym}}$ contains only sets that are symmetric with respect to the exchange of the order of $\omega_1$ and $\omega_2$. In which situations does it make sense to use $\mathscr{F}_{\mathrm{sym}}$ to model a probability space? Give a concrete example.

(b) Show that $\mathscr{F}_{\mathrm{sym}}$ is indeed a $\sigma$-algebra.

(c) Consider the two subsets of $\Omega$:

$$\begin{aligned} A &= \text{“one die shows the number 3”}, \\ B &= \text{“the blue die shows the number 3”}. \end{aligned}$$

Show that $A \in \mathscr{F}_{\mathrm{sym}}$ but $B \notin \mathscr{F}_{\mathrm{sym}}$.

(d) Consider the two maps

$$X : \begin{cases} \Omega \to \mathbb{R} \\ (\omega_1, \omega_2) \mapsto \omega_1 \end{cases} \qquad \text{“number shown by the blue die”}$$

$$S : \begin{cases} \Omega \to \mathbb{R}, \\ (\omega_1, \omega_2) \mapsto \omega_1 + \omega_2 \end{cases} \qquad \text{“sum of numbers shown by both dice”}$$

Which of these is/are a random variable/random variables on $(\Omega, \mathscr{F}_{\mathrm{sym}})$?

(a) We can for instance imagine an observe wearing colored glasses, which prevents to distinguish the two colors. For instance, for the observer, “one die shows a 1, the other a 4” is an event, namely

$$A = \{(1, 4), (4, 1)\} \in \mathscr{F}_{\mathrm{sym}}.$$

(b) We verify the properties of a $\sigma$-algebra.

(S1) Since $(\omega_1, \omega_2) \in \Omega$ implies $(\omega_2, \omega_1) \in \Omega$ for every $(\omega_1, \omega_2) \in \Omega$, we have $\Omega \in \mathscr{F}_{\mathrm{sym}}$.

(S2) Suppose $A \in \mathscr{F}_{\mathrm{sym}}$. For every $(\omega_1, \omega_2) \in \Omega$ one then has

$$(\omega_1, \omega_2) \in A \qquad \Leftrightarrow \qquad (\omega_2, \omega_1) \in A,$$

which is equivalent to

$$(\omega_1, \omega_2) \in A^c \qquad \Leftrightarrow \qquad (\omega_2, \omega_1) \in A^c,$$

so $A^c \in \mathscr{F}_{\mathrm{sym}}$.

(S3) Suppose $(A_n)_{n \in \mathbb{N}} \subseteq \mathscr{F}_{\mathrm{sym}}$. For any $\omega \in \Omega$, we have

$$(\omega_1, \omega_2) \in \bigcup_{i=1}^{\infty} A_i \Leftrightarrow \exists i : (\omega_1, \omega_2) \in A_i \overset{A_i \in \mathscr{F}_{\mathrm{sym}}}{\Leftrightarrow} \exists i : (\omega_2, \omega_1) \in A_i \Leftrightarrow (\omega_2, \omega_1) \in \bigcup_{i=1}^{\infty} A_i.$$

It follows that $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}_{\mathrm{sym}}$.

(c) We have
$$A = \{(3,i) \,:\, i \in \{1,...,6\}\} \cup \{(i,3) \,:\, i \in \{1,...,6\}\},$$
and thus $(\omega_1,\omega_2) \in A \Leftrightarrow (\omega_2,\omega_1) \in A$, for any $(\omega_1,\omega_2) \in \Omega$. It follows that $A \in \mathscr{F}_{\text{sym}}$. On the other hand,
$$B = \{(3,i) \,:\, i \in \{1,...,6\}\},$$
with $(3,1) \in B$ but $(1,3) \notin B$, so $B \notin \mathscr{F}_{\text{sym}}$.

(d) Recall that $Z : \Omega \to \mathbb{R}$ is a random $Z$ variable if $\{Z \leq a\} \in \mathscr{F}_{\text{sym}}$ for all $a \in \mathbb{R}$.
$X$ is not a random variable. Indeed, otherwise we would have
$$\{(\omega_1,\omega_2) \in \Omega \,:\, X(\omega_1,\omega_2) \leq 1\} = \{(i,1) \,:\, i \in \{1,...,6\}\} \notin \mathscr{F}_{\text{sym}}$$
(the last observation is analogous to (c)). So $X$ is not a random variable on $(\Omega, \mathscr{F}_{\text{sym}})$.
*Remark: One may also alternatively say, $X$ is a random variable on $(\Omega, \mathscr{P}(\Omega))$ but not $\mathscr{F}_{\text{sym}}$-$\mathscr{B}(\mathbb{R})$-measurable.*
We observe that
$$\{(\omega_1,\omega_2) \,:\, S(\omega_1,\omega_2) \leq a\} = \{(\omega_1,\omega_2) \in \Omega \,:\, \omega_1 + \omega_2 \leq a\} \in \mathscr{F}_{\text{sym}},$$
since $\omega_1 + \omega_2 \leq a \Leftrightarrow \omega_2 + \omega_1 \leq a$. Thus, $S$ is a random variable on $(\Omega, \mathscr{F}_{\text{sym}})$.

2. **Calculations with probability density functions**

Consider spherical bubbles of radius $R$ that form randomly in an aerated aquarium. Assume first that the radius (measured in mm) of a bubble follows a uniform distribution $R \sim \mathscr{U}([1,10])$.

(a) Calculate the probability density function of the random volume of a bubble.

For the rest of the assignment, we assume that the radius follows a *lognormal* distribution with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, $R \sim \ell\mathscr{N}(\mu,\sigma^2)$, which by definition fulfills
$$R \overset{d}{=} \exp(Y), \qquad Y \sim \mathscr{N}(\mu,\sigma^2).$$

(b) Find the probability density function of the law of $R$, that is find the probability density function of $\ell\mathscr{N}(\mu,\sigma^2)$.

(c) Show that if $R$ is lognormal distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then the volume $V$ and the surface area $A$ of the bubble are also lognormal-distributed, and find the corresponding parameters in terms of $\mu$ and $\sigma$.

(a) The cumulative distribution function of $V$ is given by $F_V(x) = \mathbf{P}[V \leq x] = \mathbf{P}\left[\frac{4\pi R^3}{3} \leq x\right] = F_R\big((\frac{3x}{4\pi})^{\frac{1}{3}}\big)$, so
$$F_V(x) = \begin{cases} 0, & \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} < 1, \\ \frac{1}{9}\left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} - \frac{1}{9}, & 1 \leq \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} < 10, \\ 1, & 10 \leq \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}}. \end{cases}$$
We obtain the probability density function of $V$ by taking the derivative:
$$\begin{aligned} f_V(x) &= \begin{cases} 0, & \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} < 1, \\ \left(\frac{3}{4\pi}\right)^{\frac{1}{3}} \cdot \frac{1}{27x^{\frac{2}{3}}}, & 1 \leq \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} < 10, \\ 0, & 10 \leq \left(\frac{3x}{4\pi}\right)^{\frac{1}{3}} \end{cases} \\ &= \begin{cases} 0, & x < \frac{4\pi}{3}, \\ \left(\frac{3}{4\pi}\right)^{\frac{1}{3}} \cdot \frac{1}{27x^{\frac{2}{3}}}, & \frac{4\pi}{3} \leq x < \frac{4000\pi}{3}, \\ 0, & \frac{4000\pi}{3} \leq x. \end{cases} \end{aligned}$$

(b) We argue similarly as in the previous subexercise. Now

$$\begin{aligned} F_R(x) &= \mathbf{P}[R \leq x] = \mathbf{P}[\exp(Y) \leq x] = \mathbf{P}[Y \leq \log(x)] \\ &= \int_{-\infty}^{\log(x)} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \mathrm{d}t, \qquad x > 0, \end{aligned}$$

and $F_R(x) = 0$ if $x \leq 0$. Again we take the derivative to find the probability density function

$$f_R(x) = \frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right) \mathbb{1}_{(0,\infty)}(x).$$

(c) Suppose that $R \sim \ell\mathcal{N}(\mu, \sigma^2)$, then we have $\log(R) \overset{d}{=} Y \sim \mathcal{N}(\mu, \sigma^2)$. For $V = \frac{4\pi R^3}{3}$, we have $\log(V) = \log\left(\frac{4\pi}{3}\right) + 3\log(R) \sim \mathcal{N}(3\mu + \log(\frac{4\pi}{3}), 9\sigma^2)$. This means that $V \sim \ell\mathcal{N}(3\mu + \log(\frac{4\pi}{3}), 9\sigma^2)$. Similarly, $A = 4\pi R^2$, so $\log(A) = \log(4\pi) + 2\log(R) \sim \mathcal{N}(\log(4\pi) + 2\mu, 4\sigma^2)$.

3. **Cumulative distribution functions**

Let $X$ be a real random variable with cumulative distribution function

$$F_X(x) = \begin{cases} \frac{1}{4}e^{x+2} + c, & x < -2, \\ \frac{1}{2} + \frac{x}{8}, & -2 \leq x < 0, \\ \frac{2+x}{3+x}, & 0 \leq x < 5, \\ (\lambda - 1)e^x + 1, & x \geq 5. \end{cases}$$

(a) Determine the values of the real constants $c$, $\lambda$.

(b) Calculate $\mathbf{P}[0 \leq X < 5]$ and $\mathbf{P}[-2 < X \leq 5]$.

Calculate the cumulative distribution function for random variables with the following distributions and draw a sketch for each:

(c) $X \sim Bin(4, \frac{1}{4})$,

(d) $Y \sim \mathrm{Cau}(1, 2)$, where $\mathrm{Cau}(\mu, a)$ stands for the Cauchy distribution with parameters $\mu \in \mathbb{R}$ and $a > 0$, characterized by the density

$$f_Y(y) = \frac{a}{\pi(a^2 + (y-\mu)^2)}, \qquad y \in \mathbb{R}.$$

*Hint:* From Calculus, it is known that $\arctan'(x) = \frac{1}{1+x^2}$, where $\arctan$ (sometimes written as $\tan^{-1}$) is the inverse of the tangent function $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$.

(a) Since $F_X$ is a cumulative distribution function, $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to\infty} F_X(x) = 1$. It follows that

$$c = 0, \qquad \lambda = 1.$$

It is also clear that $F_X$ is right-continuous and increasing (this is not a part of the assignment).

(b) It holds that:

$$\mathbf{P}[X < x] = \lim_{y \uparrow x} \mathbf{P}[X \leq y] = \lim_{y \uparrow x} F_X(y) =: F_X(x-).$$

We see that

$$\begin{aligned} \mathbf{P}[0 \leq X < 5] &= \mathbf{P}[X < 5] - \mathbf{P}[X < 0] = F_X(5-) - F_X(0-) \\ &= \frac{2+5}{3+5} - \left(\frac{1}{2} + \frac{0}{8}\right) = \frac{3}{8}. \end{aligned}$$

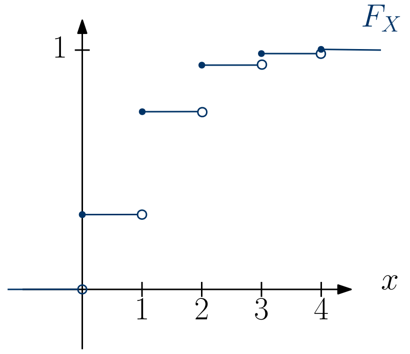
On the other hand:

$$\begin{aligned} \mathbf{P}[-2 < X \leq 5] &= \mathbf{P}[X \leq 5] - \mathbf{P}[X \leq -2] = F_X(5) - F_X(-2) \\ &= 1 - \left(\frac{1}{2} - \frac{2}{8}\right) = \frac{3}{4}. \end{aligned}$$

(c) For $X \sim Bin(4, \frac{1}{4})$, we have

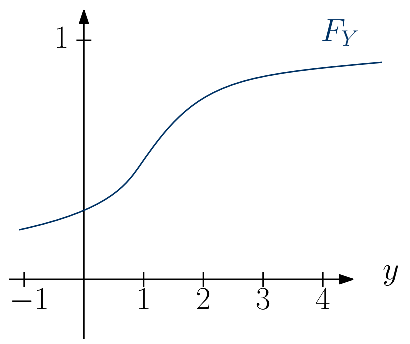
$$F_X(x) = \mathbf{P}[X \leq x] = \sum_{0 \leq k \leq \min\{4,x\}} \binom{4}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{4-k}$$

$$= \begin{cases} 0, & x < 0, \\ 0.316, & x \in [0,1), \\ 0.738, & x \in [1,2), \\ 0.949, & x \in [2,3), \\ 0.996, & x \in [3,4), \\ 1, & x \geq 4. \end{cases}$$




(d) For $Y \sim Cau(1, 2)$, we have

$$\begin{aligned} F_Y(y) = \mathbf{P}[Y \leq y] &= \int_{-\infty}^{y} \frac{2}{\pi(4 + (t-1)^2)} \mathrm{d}t \\ &= \frac{1}{2\pi} \int_{-\infty}^{y} \frac{1}{1 + (\frac{t-1}{2})^2} \mathrm{d}t \\ &= \frac{1}{\pi} \left[ \arctan\left(\frac{t-1}{2}\right) \right]_{-\infty}^{y} \\ &= \frac{1}{\pi} \left( \arctan\left(\frac{y-1}{2}\right) + \frac{\pi}{2} \right). \end{aligned}$$



4. **($*$) Medians of distribution functions** [4 Points]

A real number $m$ is called a *median* of the distribution function $F$ whenever

$$(\star) \qquad \lim_{y\uparrow m} F(y) \le \frac{1}{2} \le F(m).$$

(a) Show that $(\star)$ is equivalent to the requirement $\mathbf{P}\big[[m,\infty)\big] \ge \frac{1}{2}$ and $\mathbf{P}\big[(-\infty,m]\big] \ge \frac{1}{2}$ for the probability measure $\mathbf{P}$ associated with $F$.

(b) Show that every distribution function $F$ has at least one median.
*Hint:* Consider $\alpha = \sup\{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$, show that $\alpha$ satisfies $(\star)$.

(c) Show that the set of medians of $F$ is a closed interval of $\mathbb{R}$.
*Hint:* Consider $\beta = \sup\{x \in \mathbb{R} : F(x) \le \frac{1}{2}\}$, show that the median fulfills $m \in [\alpha, \beta]$.

(d) Show that $F(m) = \frac{1}{2}$ for any median $m$ if $F$ is continuous.

(a) Note that $\mathbf{P}[(-\infty,m]] = F(m)$, therefore

$$\mathbf{P}\big[(-\infty,m]\big] \ge \frac{1}{2} \qquad \Leftrightarrow \qquad F(m) \ge \frac{1}{2}.$$

On the other hand, $\mathbf{P}[[m,\infty)] = 1 - \mathbf{P}[(-\infty,m)] = 1 - \lim_{y\uparrow m} F(y)$. Therefore,

$$\mathbf{P}\big[[m,\infty)\big] \ge \frac{1}{2} \qquad \Leftrightarrow \qquad 1 - \lim_{y\uparrow m} F(y) \ge \frac{1}{2} \qquad \Leftrightarrow \qquad \lim_{y\uparrow m} F(y) \le \frac{1}{2}.$$

(b) Let $A = \{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$, $\alpha = \sup A$. $A \neq \emptyset$ since $\lim_{x\to-\infty} F(x) = 0$, and $\lim_{x\to\infty} F(x) = 1$.
For $\lim_{y\uparrow\alpha} F(y) \le \frac{1}{2}$: take a sequence $(\alpha_n)_{n\ge1} \in A$ with $(\alpha_n)_{n\ge1} \uparrow \alpha$, since $F$ is non-decreasing, we have

$$F(\alpha_n) < \frac{1}{2} \implies \lim_{n} F(\alpha_n) = \lim_{y\uparrow\alpha} F(y) \le \frac{1}{2}.$$

For $F(\alpha) \ge \frac{1}{2}$: suppose not, $i.e.$, $F(\alpha) < \frac{1}{2}$, since $F$ is right continuous at $\alpha$, there exist $\varepsilon > 0$, for all $x \in (\alpha, \alpha+\varepsilon)$ such that $F(x) < \frac{1}{2}$, but then $x$ must belongs to $A$, which contradicts definition of $\alpha$.

(c) Denote $\mathrm{Med}(F)$ the medians of $F$. We will show $\mathrm{Med}(F) = [\alpha,\beta]$. (i) $\mathrm{Med}(F) \subseteq [\alpha,\beta]$: If $m$ is a median, then $F(m) \ge \frac{1}{2}$, hence $m \notin A$, and $m \ge \sup A = \alpha$. Also, $F(m-) \le \frac{1}{2}$. If $m > \beta$, choose $\varepsilon = \frac{m-\beta}{2}$, we must have $F(\beta+\varepsilon) > \frac{1}{2}$, but since $F$ is non-decreasing, then $F(m-) \ge F(\beta+\varepsilon) > \frac{1}{2}$, a contradiction. Hence, $m \le \beta$. (ii) Let $m \in [\alpha,\beta]$. If $F(m) < \frac{1}{2}$, then $m \in A$, $\alpha = \sup A \ge m$ a contradiction. [Note that $m = \alpha$ should also be excluded since $F(\alpha) \ge \frac{1}{2}$ from (b). ] If $F(m-) > \frac{1}{2}$, then there exists $y < m$ with $F(y) > \frac{1}{2}$. Then for any $x \ge y$, we must have $F(x) > \frac{1}{2}$, equivalently, $x \notin B$, $B \subset (-\infty, y)$. Hence, $\beta = \sup B \le y < m \le \beta$, a contradiction, $i.e.$, $(\star)$ holds, this implies $[\alpha,\beta] \subseteq \mathrm{Med}(F)$.

(d) Since $F$ is continuous, we have

$$F(m) = \lim_{y\uparrow m} F(y) \le \frac{1}{2} \le F(m).$$